جامعة الأميرة
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كلية علوم الحاسب والمعلومات

Final report

Book Hospital Appointments

Heal Hospital Application

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Abstract

In this project, we introduce the design and evaluation of the Heal Hospital, a mobile application designed to solve issues in the hospital appointment booking process. Issues with the present appointment booking system include inefficiencies in booking, communication, and flow management, especially for old patients who face difficulties with booking appointments. The application offers a user-friendly interface that enables patients to make, reschedule, cancel appointments, get reminded about appointments, as well as view waiting time estimates. Staff at the hospital will be aided by real-time scheduling and flow management features. The system was developed using the user-centered design framework facilitated by the use of interviews, surveys, and the creation of personas. A prototype of the app was constructed and tested using manual usability testing involving five participants, and AI-based usability evaluation to prove its usability and efficiency in the field of human-computer interaction.

Introduction:

In this project, we are designing a hospital appointment booking application to help patients book their appointments more easily and reduce long waiting times. Hospitals deal with many patients every day, which makes appointment management difficult [1]. Many patients, especially elderly patients, find current booking applications confusing and stressful, and they are often unsure about their waiting time, which negatively affects their healthcare experience [2]. This increases stress for patients and pressure on hospital staff. Our goal is to create a simple and easy to use application that improves communication, reduces waiting time, and makes the hospital experience better for both patients and staff [3].

Project Problem

Hospitals deal with a large number of patients in clinics every day, leading to long waiting times and inefficiencies in scheduling systems [1]. Elderly patients often feel confused about their appointment times, which increases stress and dissatisfaction [2]. Reception and administrative staff also face challenges in managing patient flow using current systems, leading to miscommunication and information gaps. This situation highlights the need for a user-friendly and streamlined digital system to improve appointment management, support both patients and hospital employees, and reduce waiting times [3].

Project Purpose

The proposed system aims to optimize how patients experience appointment systems through a smart hospital patient flow management system [3].

The system will:
• Support patients in booking their appointments
• Provide patients with waiting time estimates
• Provide patients with notifications and reminders
• Assist hospital employees in better managing patient flow [1]
• Increase satisfaction among patients and medical staff [2]

Limitations

The system is currently a prototype that has not been incorporated into any hospital database or management systems in health care. Moreover, the software has only been tested with a few users, so it needs to be tested further with other users to ascertain its functionality and usability.

Background

Various research works have been done to tackle the problem of hospital appointments and patient flow. Digital booking systems have been shown to minimize waiting periods and increase patient satisfaction compared to manual methods [1]. Ineffective scheduling causes high levels of stress among patients and medical practitioners and results in missed appointments and clinic congestion [2].

Several existing systems have attempted to address these challenges. Sehhaty, a widely used healthcare application in Saudi Arabia, provides online booking services and digital health records, minimizing the need to visit the clinic in person and improving access to healthcare services [3]. Its strengths include remote accessibility and ease of booking. However, it lacks live queue tracking and precise waiting time updates, which causes uncertainty and anxiety for patients after they arrive at the clinic.

Zocdoc, a widely used appointment booking platform, allows patients to search for doctors by specialty, location, and availability, and book appointments instantly online. Its strengths include a clean interface, real-time availability updates, and patient reviews for doctors [6]. However, Zocdoc is primarily designed for tech-savvy users and does not offer waiting time tracking inside the clinic or sufficient support for elderly patients who require simpler navigation [2].

Similarly, general hospital management systems focus primarily on administrative and organizational efficiency. Their strength lies in supporting staff with scheduling and record management. However, they are often not designed with the patient experience in mind, making them difficult to use for elderly patients or those with limited technical experience [2].

The Human-Computer Interaction field emphasizes the significance of user-centered design that prioritizes user needs and simplicity, particularly in healthcare applications [4]. Moreover, reminders and notifications have been shown to be effective in reducing missed appointments and improving patient engagement [5].

The Heal Hospital application addresses the weaknesses identified in existing systems by combining real-time queue tracking, appointment reminders, and a simple user-centered interface accessible to all user types, including elderly patients.

Analysis:

The transition toward digital healthcare is a necessity, as evidenced by our multimethod analysis. Our investigation, including field interviews conducted by Team member 1 and Team member 2, reveals a system under pressure. While relying on manual coordination, our survey of staff—77.8% of whom are receptionists like Aisha — highlights that existing workflows cause operational strain. A significant 44.4% reported that the system increases workload, with 77.8% identifying "System Slowness" as their persistent daily challenge. This inefficiency ripples down to patients, as 66.7% of staff confirm that prolonged waiting times disrupt reception duties.

Despite these bottlenecks, several positive practices offer a foundation for improvement. A key strength is the high digital readiness; 88.9% of staff use the system daily, ensuring a smooth transition. Furthermore, the habit of following structured appointment lists provides organizational discipline. As noted by Team member 1 in her interview with Abdullah, the dedication to these lists is clear, yet manual coordination remains the primary failure point.

Existing systems like "Sehhaty" successfully introduce remote booking, but a "localized gap" remains in managing live clinic flow. This lack of transparency causes "uncertainty anxiety," captured by team member 1

during her interview with Fatimah, an elderly patient. Similarly, interviews by Team member 2 with Dr. Turki and Nurse Rose emphasized that overlapping schedules disrupt workflows. Physiotherapist Norah also highlighted the "chaos" caused by fragmented communication. By addressing the 77.8% of staff requesting improved usability, our solution will ensure a streamlined experience for professionals like Khalid Alharbi and patients alike.

1. Users:

For a hospital appointment system, the primary and secondary users and their characteristics can be outlined as follows:

Primary User (Internal User)

1. Hospital Flow Manager

Hospital Flow Managers are internal users responsible for monitoring patient flow in outpatient clinics. They use the system to organize schedules, reduce waiting times, and improve operational efficiency.

Some of the Hospital Flow Manager's characteristics:

1.1. Organization-oriented: Managers utilize the system to coordinate patient appointments and clinic schedules.
1.2. Efficiency-focused: They rely on real-time updates to prevent overcrowding and reduce delays.
1.3. Decision-support: Managers use the system to monitor patient flow data and make informed operational decisions

Persona 1 – Primary User

Khalid is a 31-year-old hospital flow manager in Riyadh. He has a degree in Health Administration. He oversees patient queues and appointments. He also coordinates the clinic staff which requires him to use technology and digital dashboards with real-time updates. Booking Overcrowding and appointment pushes are major operational issues to him. He is seeking a system that can assist him in tracking wait times and booking pushes. He hopes to reduce the backlogs and ease the flow for patients and clinic staff.



2.Reception Staff (Hospital Receptionist)

Reception staff are internal users who interact directly with patients and manage daily appointment-related tasks. They are responsible for booking appointments, updating schedules, answering patient inquiries, and supporting patient flow at the reception desk. Receptionists work in fast-paced environments and rely heavily on the appointment management system to reduce manual work, avoid errors, and improve patient experience.

Some of the Reception Staff Characteristics:

2.1 Task-oriented:
Receptionists focus on completing daily operational tasks such as registering patients, booking appointments, and confirming appointment details.

2.2 Efficiency-focused:
They require a fast and reliable system to handle high patient volumes, especially during peak hours.

2.3 Communication-focused:
Receptionists act as a link between patients, doctors, and hospital departments, ensuring accurate information exchange.

Persona 2 – Primary User (Reception Staff)



Name: Aisha
Age: 29
Occupation: Hospital Receptionist
Location: Riyadh

Aisha works as a receptionist in a public hospital. She is responsible for welcoming patients, booking and modifying appointments, and answering patient questions. She often deals with elderly patients who need extra assistance during the booking process. During busy clinic hours, she experiences stress due to long queues and frequent appointment changes. Aisha prefers a system that is easy to use, reduces manual data entry, and provides quick access to appointment schedules. Her goal is to manage patient appointments efficiently while reducing waiting times and patient confusion

3. Doctor/Nurse/Physiotherapist

Doctors and nurses and physiotherapists are internal users who rely on the system to track patient appointments and manage daily clinical schedules therapy sessions. They use the system to monitor patient flow, reduce delays, and ensure that appointments therapy sessions are organized efficiently during clinic hours.

Some of the healthcare providers characteristics:



3.1. Workflow-oriented:
Healthcare providers use the system to follow daily appointment schedules and manage patient flow effectively.

3.2. Efficiency-focused:
They rely on clear schedules and real-time updates to reduce delays, avoid overcrowding, and manage their time efficiently.

3.3. Coordination-support:
Healthcare providers depend on the system to coordinate with reception staff and stay informed about appointment changes or delays .

Persona 3 – Primary User (Doctor/Nurse)

Dr. Aisha is a 38-year-old internal medicine physician in Riyadh who manages a busy daily schedule with many patients. She relies on digital systems to track appointments, monitor patient flow, and stay updated on delays. Due to limited time and high workload, she prefers clear and efficient interfaces with real-time updates. She hopes the system will help reduce waiting times, improve coordination with staff, and minimize administrative stress.



Persona 4 – Primary User (Physiotherapist)

Name: Norah
Age: 27
Occupation: Physiotherapist
Location: Riyadh

Norah is a physiotherapist working in the hospital rehabilitation department. She conducts multiple therapy sessions each day and the success of her treatment plans relies heavily on her ability to schedule her day correctly. She often liaises with the reception team to manage her patient bookings and reschedule therapy sessions. Her work flow gets interrupted when patients show up for therapy without appointments. Last minute therapy appointment changes can often create chaos. If therapy sessions overlap or are scheduled at the same time, it increases stress to Norah and the team. The therapy sessions that are scheduled after the delayed session will end up creating longer wait times for the patients.

As a Physiotherapist, Norah needs a clear and structured system to provide her with a daily planner and real-time update. She prefers a clear and organized system to manage her appointments, conflicts and time. Her goal is to deliver uninterrupted, high-quality therapy sessions while minimizing delays.

Secondary User (External User)

1. Elderly Patients

Elderly patients are external users who visit the hospital for medical consultations. They interact with the system to book appointments, receive reminders, and check waiting times.

Some of the Elderly Patient's characteristics:

1.1. Patient-oriented: Patients use the system to schedule medical appointments and track their waiting times.
1.2. Simplicity-focused: They prefer a clear, easy-to-navigate interface suitable for low technical skill levels.
1.3. Information-seeking: Patients rely on the system for reminders, notifications, and updates about their appointments.

Persona 1 – Secondary User

Fatimah is a 55-year-old retired teacher in Riyadh. She regularly visits hospitals for check ups and to monitor chronic conditions. She is not very good with technology and uses her smartphone only to call and message. She typically asks family members for help because booking appointments is often confusing to her. She gets anxious and exhausted from long waits and uncertainty. She appreciates simple interfaces with small step instructions, large text, and reminder notifications. She hopes to be able to book appointments and know the expected wait time for appointments before arriving.



2.Family Caregiver / Patient Companion

General Description:

The Family Caregiver is an external user who assists patients—especially elderly or dependent patients—with booking appointments, receiving reminders, and understanding medical visit details. This user does not manage the system or make administrative decisions but uses the system to support the patient.

Some of the Family Caregiver characteristics:

2.1 Support-oriented: Uses the system to assist a family member in booking appointments and tracking visit details.
2.2 Time-sensitive: Focused on managing appointment times and waiting durations to avoid delays.
2.3 Information-focused: Relies on the system for clear, accurate information about appointments, clinics, doctors, and notifications.

Persona 2– Secondary User



Name: Aisha
Age: 28
Occupation: University Student
Location: Riyadh

Aisha often accompanies her mother to hospital appointments. Since her mother finds digital systems difficult to use, Aisha handles appointment bookings, receives reminders, and monitors waiting times on her behalf. She prefers systems that are clear, fast, and accessible via mobile devices. Notifications and reminders are essential for her to manage time efficiently and ensure that her mother's hospital visits are organized, smooth, and stress-free

3. Patients

Patients are external users who interact with the system by receiving appointment notifications, reminders, and updates. They benefit from improved appointment organization and reduced waiting times as a result of better system support for medical staff.

Some of the Patient characteristics:

3.1. Information-oriented:
Patients rely on the system to receive clear information about appointment times, delays, and reminders.
3.2. Notification-dependent:
They depend on notifications and reminders to avoid missing appointments and to stay informed about any changes.
3.3. Experience-focused:

Patients expect a smoother appointment experience with reduced waiting times and better communication.



Persona 3 – Secondary User (Patient)

Ahmed is a 45-year-old office employee in Riyadh who regularly visits hospitals for follow-up appointments. He uses his smartphone daily but feels frustrated by unclear systems and long waiting times. He depends on reminders and notifications to manage his appointments and values clear information about timing and delays. He hopes the system will provide better communication and a smoother, more predictable hospital experience.

4. Young Patient

Young patients are external users who rely on digital tools to manage their hospital visits. Digital appointment management systems should be quick, adjustable, and flexible, making it easy for users to book, change, or cancel appointments. Young patients mainly use mobile apps and websites.
to interact with the system. They appreciate convenience, clarity, and time management toolsSome of the Young Patient's characteristics:

4.1. Technology-centered: Young patients prefer the use of mobile apps and digitized systems for their healthcare services.

4.2. Efficiency-centered: Young patients prefer bookings that do not require a waiting period and eliminate extra steps.

4.3. Information-centered: Young patients appreciate detailed instructions and are reliant on reminders and notifications, so they do not lose track of their appointments.

Persona 4 – Secondary User (Young Patient)



Name: Raneem
Age: 18
Occupation: University Student
Location: Riyadh

Raneem is a university student who enjoys using mobile apps to simplify various tasks, including booking appointments at the hospital. She appreciates efficiency, convenience, and flexibility, and she values the ability to easily and quickly book an appointment.

Raneem becomes frustrated when she sees appointment slots that have not been updated or when she does not receive appointment reminders.

She dislikes long waiting times and unclear information about delays. Since she manages a busy academic schedule, she needs a reliable system that allows her to book, modify, or cancel appointments easily.

Her goal is to have a fast, mobile-friendly appointment system that provides real-time updates, estimated waiting times, and timely notifications. She wants a smooth and predictable hospital experience that fits into her daily routine without causing stress or confusion.

2. Tasks:

This task shows the steps that users take to complete appointments and the different ways these tasks can be done with and without the digital system.

1. Searching for a doctor

a. With System:
Patients can search for doctors and filter results based on doctor specialty, availability, and distance.

b. Without System:
Patients have to choose doctors based on word-of-mouth, which may not always be accurate.

2. Reviews and ratings of doctors

a. With System:
Patients can see ratings and reviews of other patients regarding a doctor.

b. Without System:
Patients have to choose doctors based on word-of-mouth, and the advice given is not guaranteed to be accurate.

3. Scheduling an appointment

a. With System:
Patients can make appointments through mobile apps and websites of hospitals by selecting a doctor and choosing the date and time of the appointment.

b. Without System:
Patients have to go to a hospital, which may involve a long waiting time, or call the reception of the hospital.

4. Changing and canceling appointments

a. With System:
Patients can change and cancel appointments through the system, or call the hospital and do so.

b. Without System:
Patients have to visit the hospital or make repeated calls to have those changes completed.

5. Wait Times

a. With System:
Patients can see how long they have to wait, and how many people are ahead of them.

b. Without System:
Patients wait with no idea about how long they will be waiting.

6. Appointment Reminders

a. With System:
Patients receive reminders of the appointments by push notifications, SMS messages, or phone calls.

b. Without System:
Patients have to rely on themselves to remember the appointment or write it down, resulting in them forgetting the appointment.

7. Patient Flow Management (What Receptionists Do)

a. With System:
Receptionists are able to use the system to book appointments and to see how many patients are in line.

b. Without System:
Receptionists are likely to fall behind in the appointment management workflow.

8. Managing the Schedule of the Clinician

a. With System:
Doctors and physiotherapists manage their schedules, see which patients are checked in, and can make changes to appointments.

b. Without System:
They use a paper schedule or have someone tell them, which leads to overlapping appointments and workflow disruptions.

3. Interviews, Field Study or Questionnaire results:

- **Interviews**

First Interview:

(Done by Team member 1)

Two people were interviewed, and their opinions were taken into consideration. We listened attentively to their perspectives and ensured that their feedback was reflected in our application.

The questions for Fatimah (Elderly Patient, my mother):

1. How do you usually plan your visits to the hospital?
• I usually call the hospital or visit the reception in person to check available appointment times.
2. How simple or complicated do you think the appointment system is?
• It is confusing and takes a lot of time to understand.
3. How long do you usually wait to see the doctor?
• I often wait more than 30–40 minutes, sometimes even longer.
4. What do you do to pass the time while you are waiting?
• I usually use my phone or talk with other patients, but it can get stressful.
5. What do you think the appointment system should include?

• A simple, easy-to-use interface, clear instructions, reminders, and notifications for appointments.

Second Interview:

(Done by Team member 1)

The questions for Abdullah (Hospital Flow Manager, my brother):

1. How do you currently monitor patient queues and manage appointments?

• I check lists manually and coordinate with the reception staff, but it's timeconsuming.

2. What challenges do you face with the current system in organizing patient flow?

• Delays, miscommunication, and overcrowding in the waiting areas.

3. What features or improvements would help you manage appointments and reduce congestion more effectively?

• A digital system that provides real-time updates, alerts when patients arrive, and organizes schedules efficiently

Summary of Findings from the Interviews

- The appointment booking process takes a long time and is confusing
- Fatimah feels stressed and are often unsure about how long the wait will be
- Fatimah would like clear instructions and an automated system to remind and notify her
- Fatimah appreciate the ability to digitally monitor their queue status
- Abdullah struggles to manage patient flow and appointments efficiently with the current system. He needs a digital solution that provides real-time updates and helps staff coordinate schedules effectively.

The system must be designed specifically for elderly patients. It should provide clear, visible, and easy-to-read information. while also supporting hospital staff in managing patient flow.

Third Interview:

(Done by Team member 2)

Two medical staff members were interviewed, and their opinions were taken into consideration. We listened attentively to their perspectives and ensured that their feedback was reflected in the proposed hospital appointment booking system.

The questions for Dr. Turki (My uncle is a doctor):

1. How do you currently manage patient appointments during your workday?

• I usually follow the appointment schedule prepared by the reception staff, but the schedule often changes during the day due to delays and unexpected cases.

2. What challenges do you face with the current appointment booking system?
• The main challenges are unclear schedules, overlapping appointments, and long patient waiting times.
3. How does long patient waiting time affect your daily work?
• Long waiting times increase stress for both patients and medical staff and make it difficult to maintain an organized workflow.
4. Do appointment delays affect patient satisfaction from your perspective?
• Yes, many patients become frustrated when they wait too long, which negatively affects their overall experience.
5. What do you think the appointment system should include to improve your work? • A digital system that provides clear schedules, real-time updates, and notifications when delays occur.

Fourth Interview:

(Done by Team member 2)

The questions for Nurse Rose (My friend is a Nurse):
1. How do you assist doctors in managing patient appointments?
• I help prepare patient lists and coordinate with reception staff, but delays often make this process difficult.
2. What difficulties do you notice when appointments are delayed?
• Patients become anxious and impatient, and staff workload increases significantly.
3. How does long waiting time affect patients and staff?
• It causes stress for patients and makes it harder for staff to manage patient flow efficiently.
4. Do reminders and notifications help reduce missed appointments?
• Yes, reminders help patients remember their appointments and arrive on time.
5. What improvements would make appointment management easier?
•A simple system that shows clear appointment times, waiting time estimates, and sends notifications would be very helpful.

Summary of Findings from the Interviews

- Doctors and nurses face difficulties managing patient appointments due to unclear schedules, frequent delays, and long waiting times.
- Long waiting times increase stress for both patients and medical staff and negatively affect workflow efficiency.
- Medical staff need a digital appointment system that provides real-time updates and clear scheduling information.
- Notifications and reminders are essential to reduce missed appointments and improve coordination between staff and patients.

The system must support doctors and nurses in managing patient flow efficiently while also improving communication with patients to reduce waiting times and increase overall satisfaction.

Fifth Interview:

(Done by Razan Alshahrani)

To tailor the app system integrated with the appointment booking system, an interview with a healthcare professional was done to clarify how the medical domain personnel deal with such systems and to identify the specific challenges faced in terms of patient traffic management.

The questions for Norah (Physiotherapist , my sister):

1. What is your current process in receiving and reviewing physiotherapy session appointments?
• Daily, I check the hospital daily schedule system, and from time to time, I get updates from the receptionist. The issue is that the reception staff are not always providing an updated version, which generates some confusion.

2. What are the challenges you face with the current appointment system when managing therapy sessions??
• The most problematic situation is when changes are not made in the system. For some patients, they don't have a confirmed appointment, and then they show up, which complicates the plan.

3. What is the effect on your work and the quality of physiotherapy sessions when patients arrive late or when sessions overlap?
• The quality of the physiotherapy session is affected when there is pressure on the staff and when multiple patients are waiting at the same time. Delays make it difficult to manage session time properly and impact the patient's progress.

4. Which characteristics would assist you in the more effective management of appointments?
• Notifications and updates about delays and cancellations are useful, but an ideal scheduling system would adjust the schedule automatically in response to any changes.

5.How significant is the communication between reception and physiotherapists? • Significant. When there is no communication, there is a delay, and patients get frustrated. A system integrated for both would lessen these problems.

6. What features would you want to be included in a new appointment management system for physiotherapy sessions?
• A dashboard that includes patient information, alerts about delays/wait times, and notifications in real time about updates. This would help avoid delays and provide better preparation on our end.

Summary of Findings from the Interviews

- The physiotherapist has a lack of updated schedules from reception, leading to confusion.
- Patients arriving with no appointment can disrupt the therapy sessions.
- Overlapping appointments reduce the quality of care and affect patient progress.
- The physiotherapist would like to have more updated schedules, more communication, and more real time tools.
- The physiotherapist is not fully satisfied with the current workflow due to the lack of real-time updates and communication.

Sixth Interview:
(Done by Razan Alshahrani)

Seeking young patient feedback on the hospital's appointment online systems, the young patient's feedback was analyzed and used to inform the system's requirements.

The questions for Raneem (Young patient):

Responses to Interview Questions

1. How do you book hospital appointments?
• I use mobile applications or hospital websites because they offer speed and convenience.

2. What booking method do you prefer? Why?
• I prefer mobile apps because they let me book whenever I want and they show me my available options.

3. How much time do you spend on booking?
• If the app is designed properly, it should not take more than a few minutes.

4. What difficulties and challenges do you encounter when booking appointments?
• The system may sometimes not update the available times.

5. Describe a time when you missed or forgot your appointments. What happened?
• That has happened to me when the system did not remind me or notify me properly.

6. What features would you want in an ideal appointment booking application?
• Ideal booking apps would include reminders, fast booking, an option to change/cancel appointments.

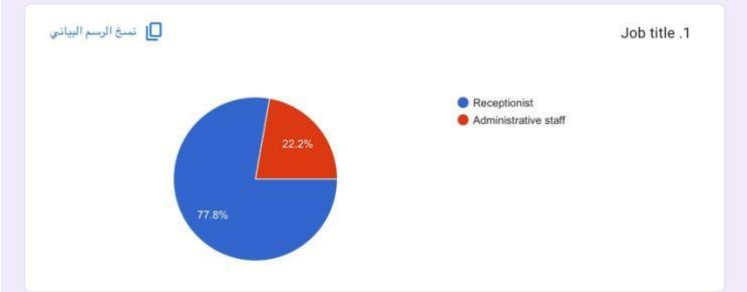
Summary of Findings from the Interviews

- The appointment system can be slow and confusing.
- The patient prefers mobile apps but has difficulties when appointment times are not updated.
- Without a reminder, an appointment might be forgotten.
- The patient wants more explicit instruction, faster booking, and more dependable reminder systems.
- The patients appreciate being able to see how many patients are ahead of them in the queue.
- Overall, the patient is dissatisfied with the current system due to outdated information and missing notifications.
- The patient believes the system needs to be more responsive, clear, and reliable.

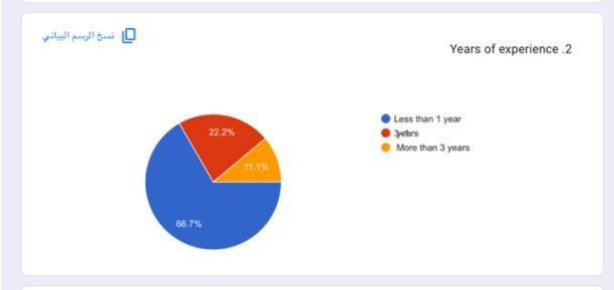
Survey:

(check the survey link)
https://docs.google.com/forms/d/e/1FAIpQLScxAdFV-QOcn_c_io_EV5_WaUg_YcNinWueix1XWZzF9HDMQw/viewform

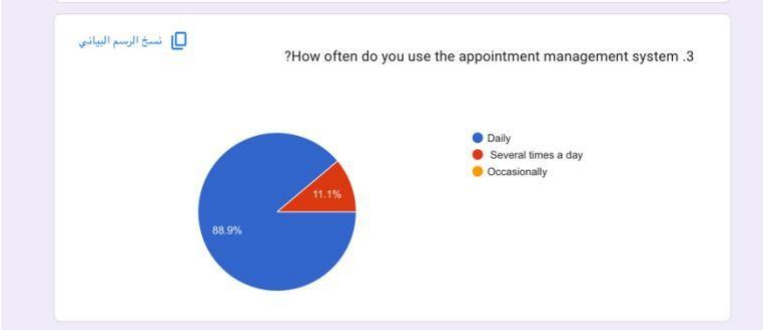
To summarize our findings:



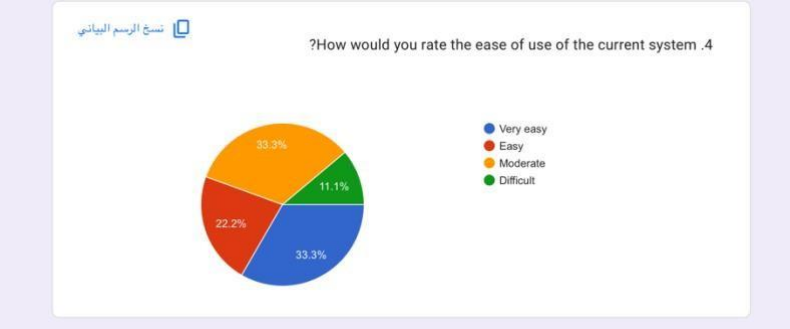
Job Title: Most participants are receptionists (77.8%), followed by administrative staff (22.2%).



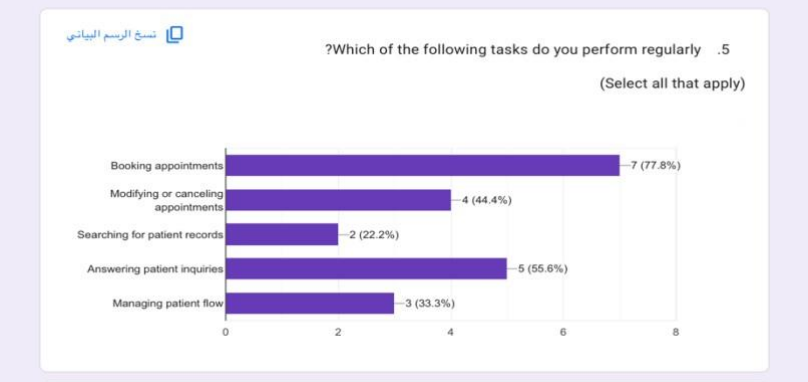
Years of Experience: A majority of the staff (66.7%) have less than one year of experience



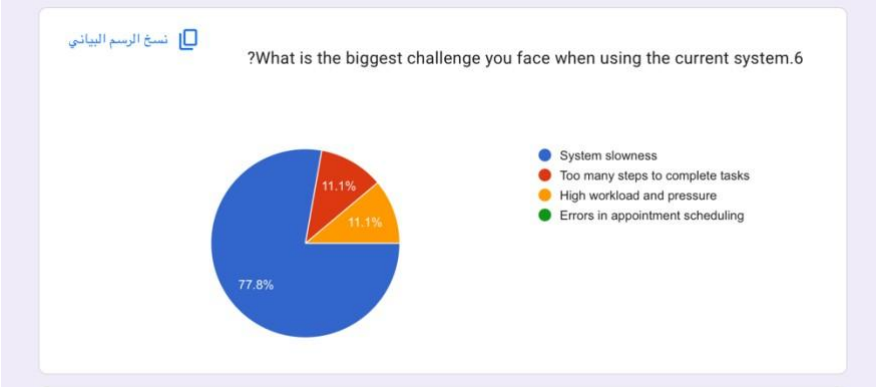
Usage Frequency: The system is used heavily, with 88.9% of staff relying on it daily.



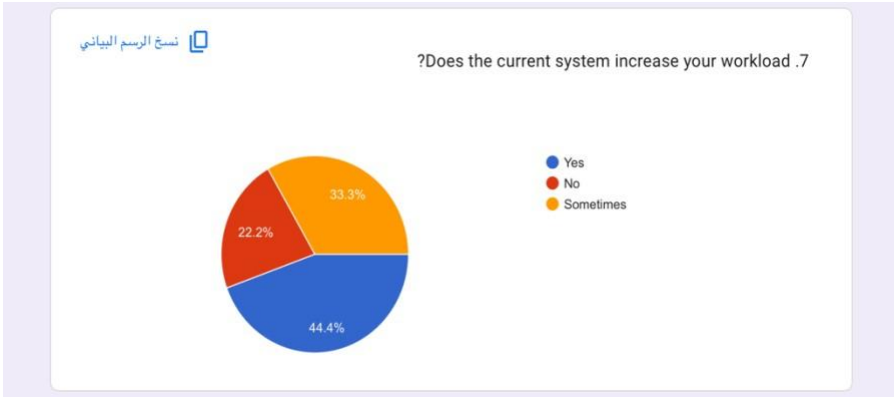
Ease of Use Rating: Satisfaction is inconsistent; only 33.3% find the current system "Very Easy" to use.



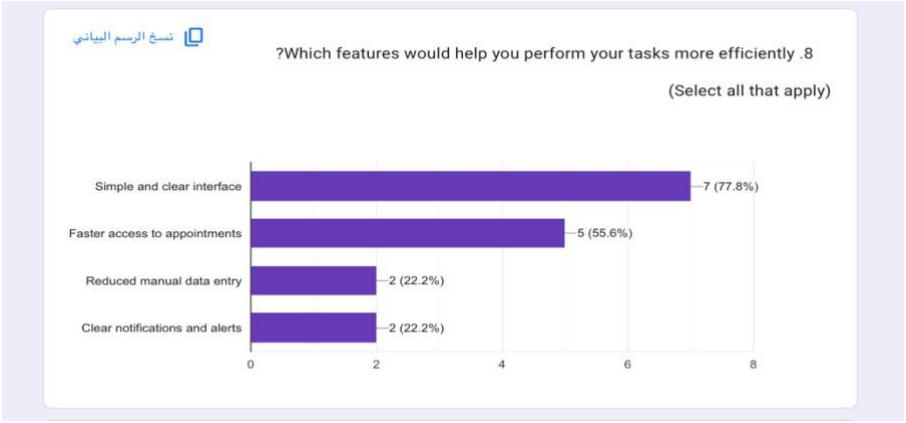
Regular Tasks: The most frequent task is "Booking Appointments" (77.8%), followed by answering inquiries (55.6%).



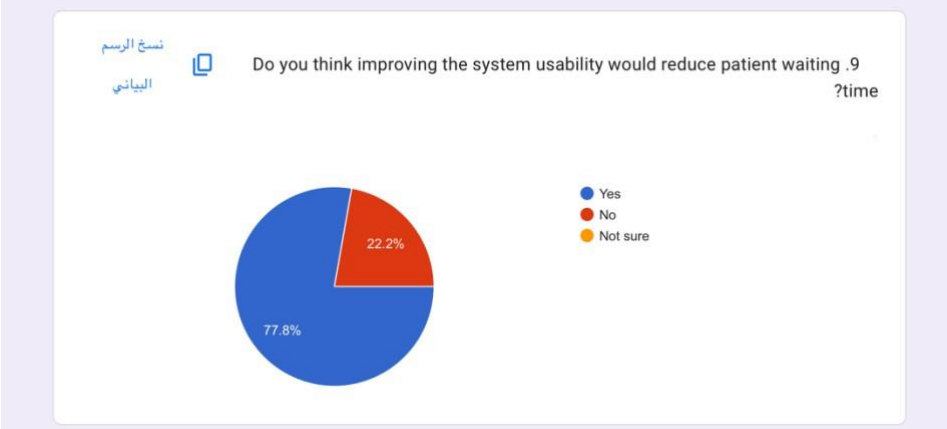
Biggest Challenge: "System Slowness" is the dominant problem, reported by 77.8% of users.



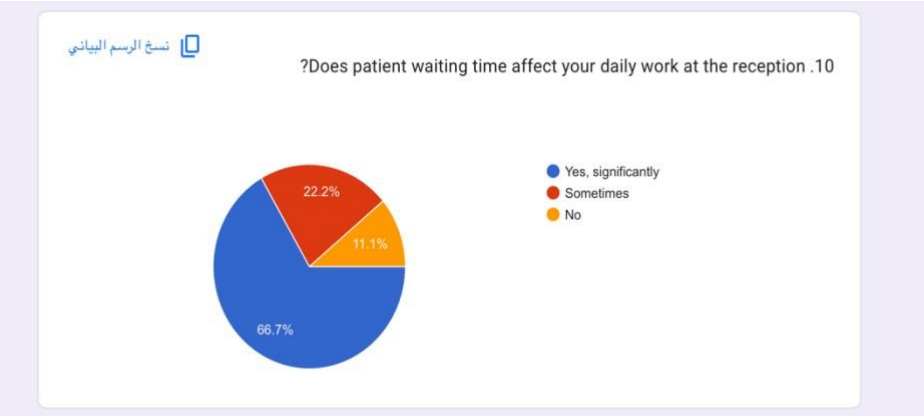
Workload Impact: 44.4% of respondents feel the current system actively increases their workload.



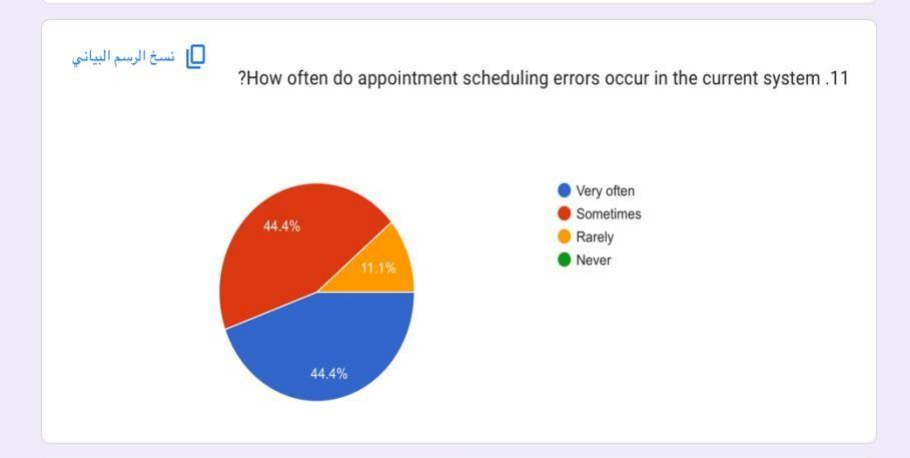
Efficiency Features: 77.8% of staff agree that a "Simple and Clear Interface" would best improve their efficiency.



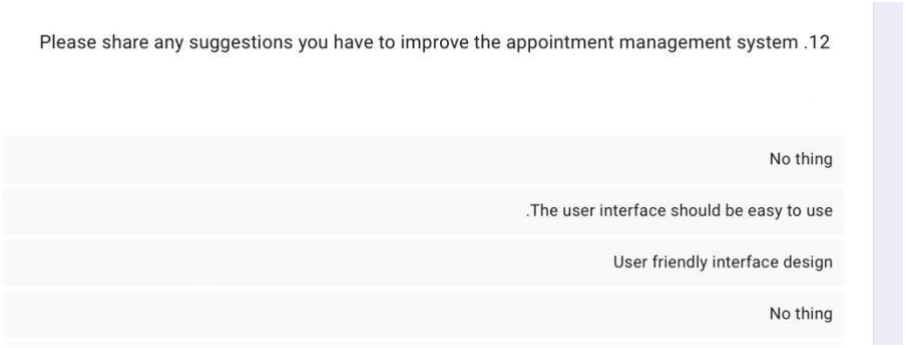
Usability & Waiting Time: 77.8% believe that better system usability would directly reduce patient waiting times.



Impact on Reception Work: 66.7% confirm that long patient waiting times significantly affect their daily productivity.



Scheduling Errors: System reliability is a major issue, with 44.4% encountering errors "Very Often."



User Suggestions: A "User-friendly" interface to improve efficiency.

4. Requirements

Functional Requirements

1. Appointment Booking and Navigation
1.1. User will be able to easily navigate the hospital appointment booking system to access booking, modification, and appointment information pages.
2.1. The system will provide a clear and simple booking interface that allows users to select a clinic, doctor, date, and time.

2. Appointment Availability and Scheduling
1.2. User will be able to view available appointment slots before confirming the booking.
2.2. The system will organize appointment schedules to reduce conflicts and overcrowding.

3. Appointment Modification and Cancellation
1.3. User will be able to modify or cancel appointments easily through the system without visiting the hospital.
2.3. The system will update appointment schedules automatically after any changes are made.

4. Waiting Time Management
1.4. The system will display the estimated waiting time for patients after booking an appointment.

2.4. The system will provide real-time updates in case of delays in the appointment schedule.

5. Notifications and Reminders

1.5. User will be able to receive notifications regarding appointment confirmations and changes.
2.5. The system will send appointment reminders to reduce missed appointments.

6. Support for Elderly Patients
1.6. The system will provide clear instructions, readable text, and simple navigation suitable for elderly patients.
2.6. The system will minimize the number of steps required to complete the booking process.

7. Staff Support and Patient Flow Management
1.7. The system will allow doctors, nurses, and reception staff to view daily appointment schedules clearly.
2.7. The system will support hospital staff in managing patient flow and reducing long waiting times.

Non-Functional Requirements

1. Usability and Accessibility
1.1. The system will be easy to use and accessible for users with different technical skill levels, especially elderly patients.
2.1. User interface elements will be clear and intuitive, allowing users to complete tasks with minimal effort.

2. Performance and Reliability
1.2. The system will respond quickly to user actions such as booking, modifying appointments, and viewing schedules.
2.2. The system will remain reliable during peak hours to support continuous appointment management without delays.

1. System logo

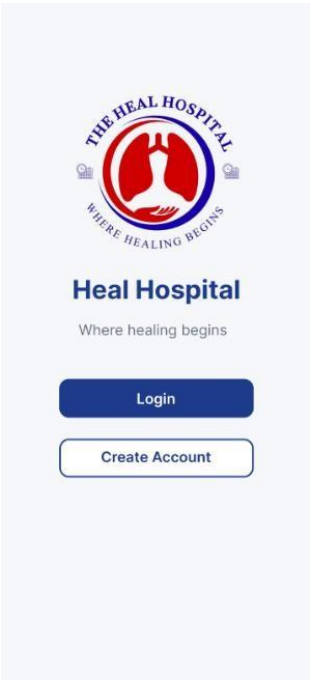


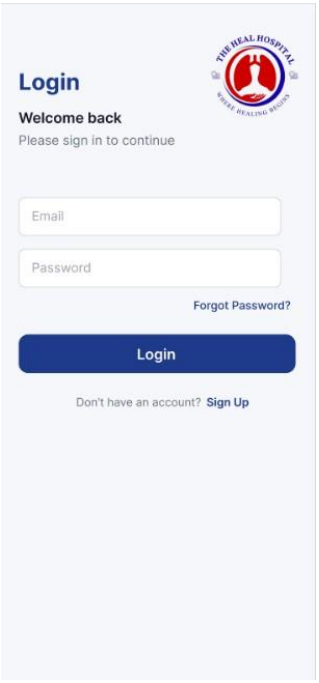
The Heal Hospital logo was made to symbolize healthcare support, the experience of the patient, and the start of the healing process. The major symbol of the logo is the lungs above the hand. The lungs signify medical treatment, health, and life. The hand represents the support and cares the hospital offers to its patients. All of these elements demonstrate the hospital's pledge to uphold the health of patients.

The ideas of organization, protection, and continuity in healthcare services are also represented in the circular shape of the logo. The Heal Hospital system aims to provide a more organized structure to enhance the hospital experience for patients. The system facilitates appointment booking, waiting time prediction, appointment reminders, and simplifies stress and confusion. Adding calendar and clock icons further clarifies the system's purpose and reinforces the ideals of structured appointment scheduling, time management, and punctual reminders which help keep patients informed and reduce uncertainty.

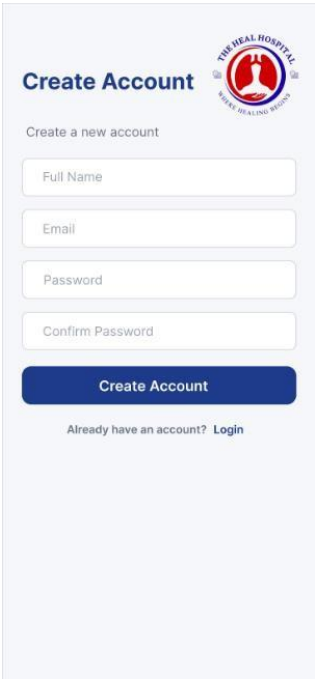
The logo also utilizes specific colors to symbolize various things. Red represents urgency, medical care, and the hospital's commitment to helping and treating patients whenever needed. Dark blue represents trust, professionalism, and reliability, which are essential to healthcare services. The tagline "Where Healing Begins" highlights that the system supports patients from the first step of their healthcare journey, starting with booking their appointment.

2. System prototype

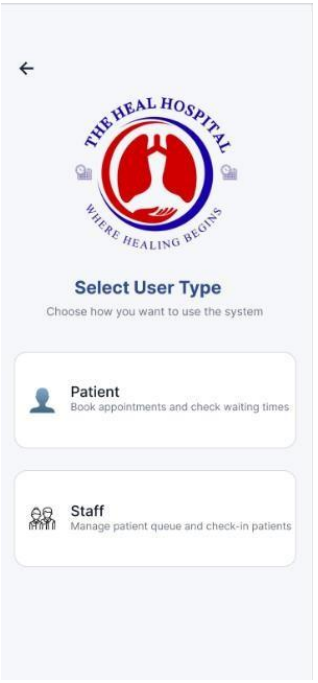


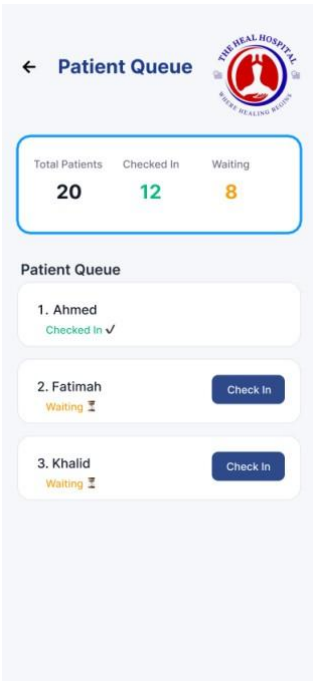


If the user clicks "on Login"

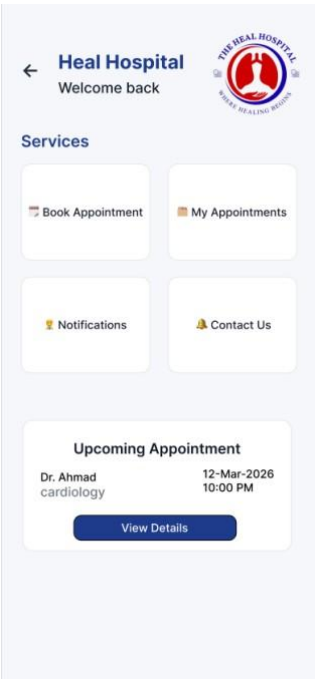


If the user clicks "Create Account"

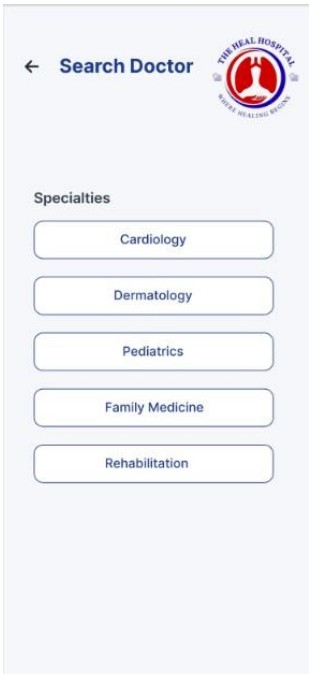




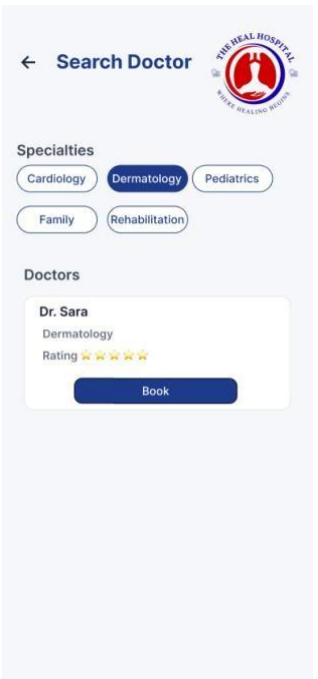
If the user type is Staff



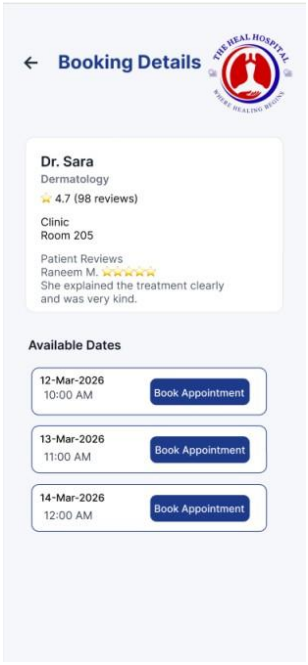
If the user type is Patient

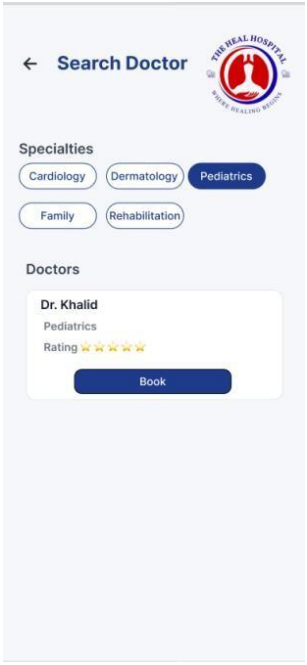


If the user clicks on "Book Appointment"

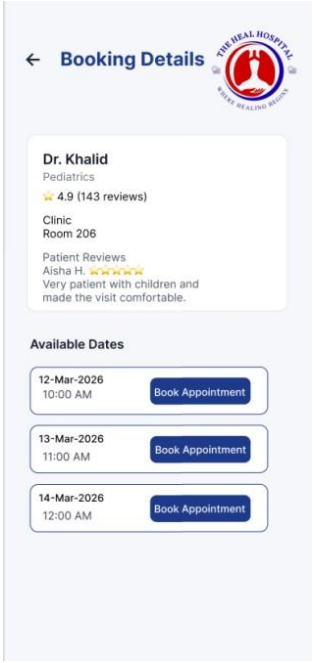


If the user clicks on "Dermatology"

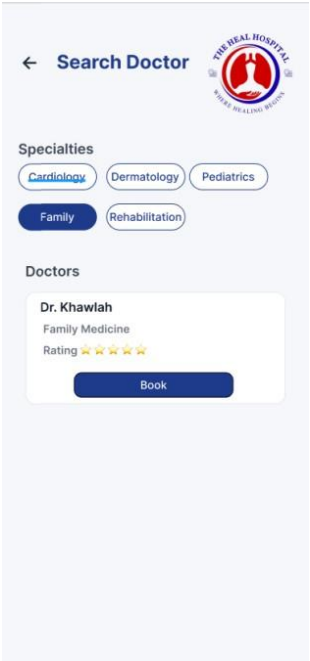




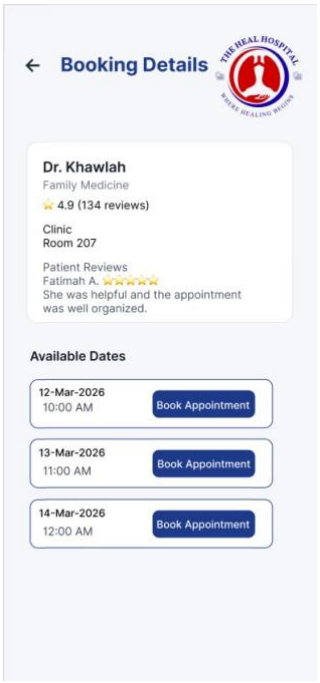
If the user clicks on "Pediatric"



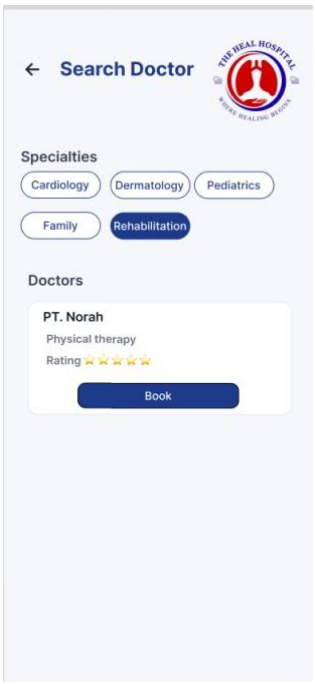
If the user clicks on "Book"



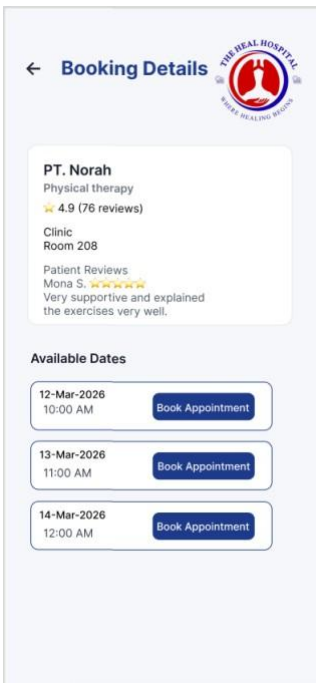
If the user clicks on "Family"



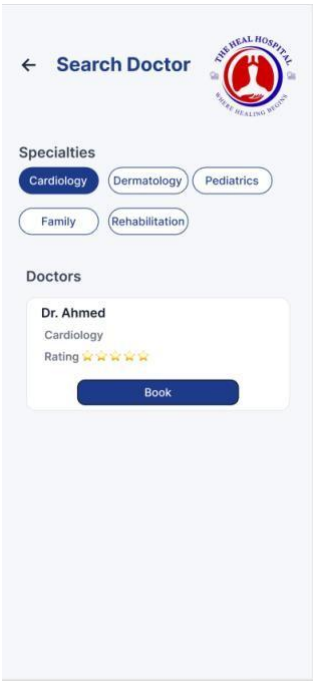
If the user clicks on "Book"



If the user clicks on "Rehabilitation"



If the user clicks on "Book"

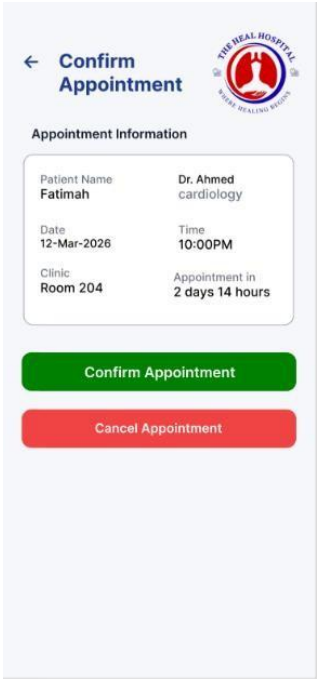


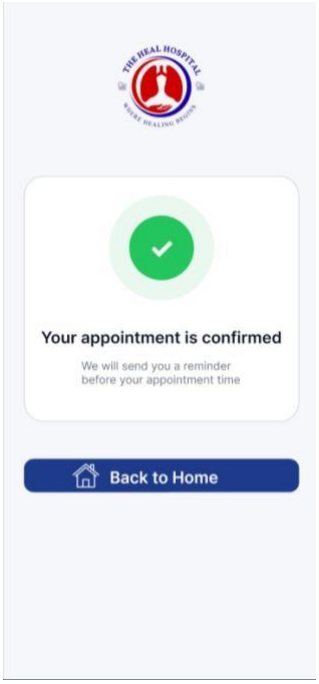
If the user clicks on "cardiology"



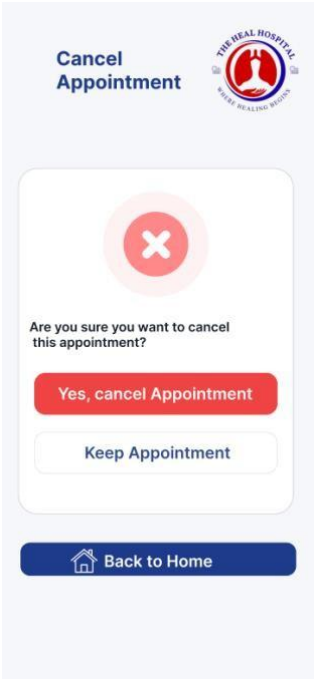
If the user clicks on "Book"

Note: The same booking flow is used for all doctors.

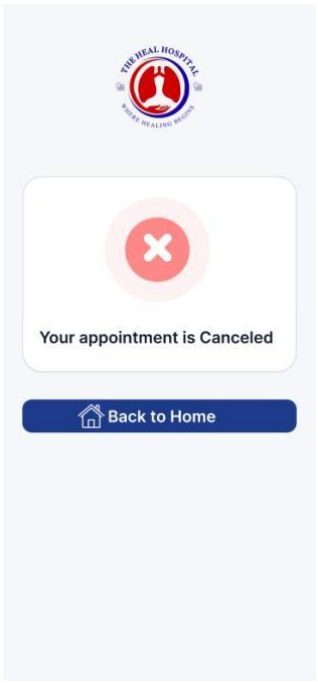




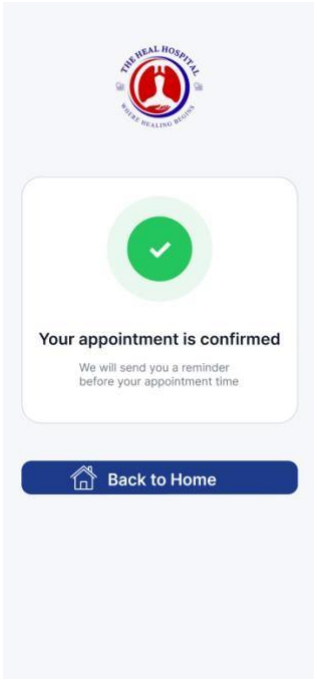
If the user clicks on "Confirm Appointment"



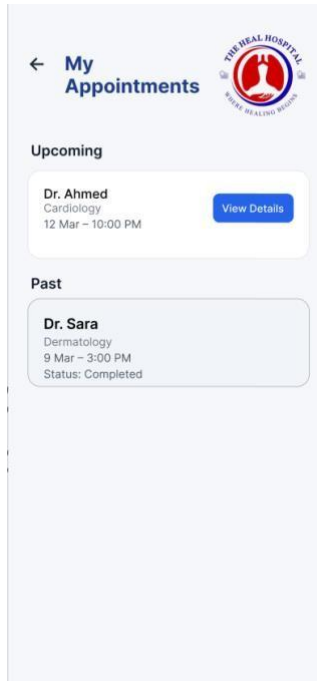
If the user clicks on "cancel Appointment"



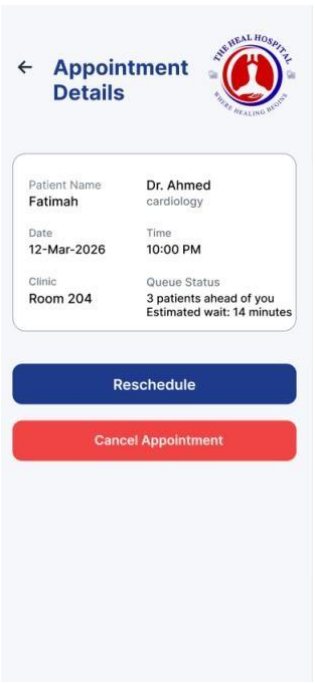
If the user clicks on "Yes, Cancel Appointment"



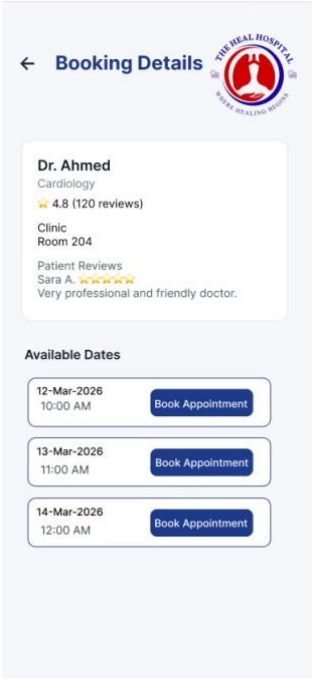
If the user clicks on " Keep Appointment"



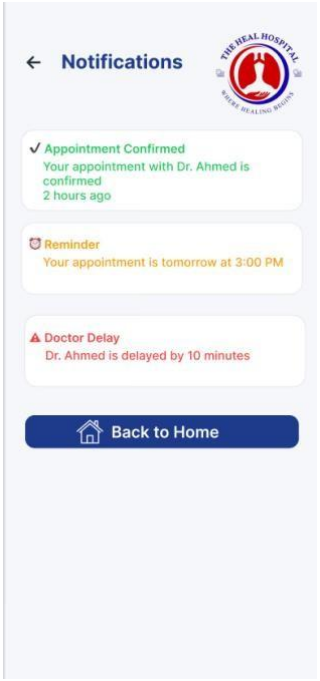
If the user clicks on "My Appointment" From Home page



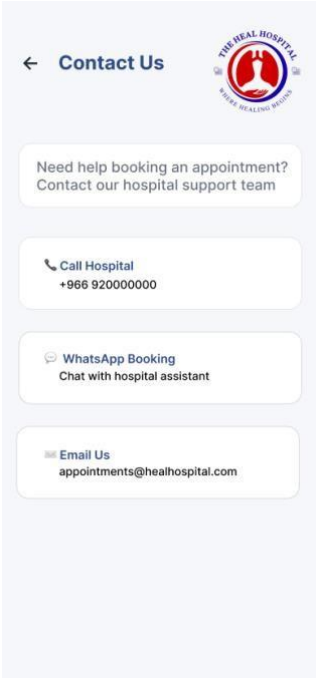
If the user clicks on "View Details "



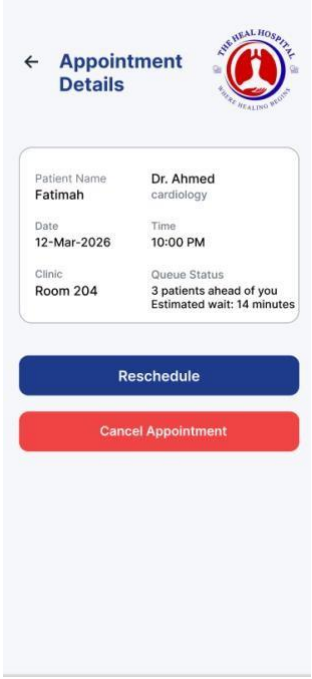
If the user clicks on "Reschedul"



If the user clicks on " Notifications" From Home page



If the user clicks on " Contact Us" From Home page



If the user clicks on " Upcoming Appointment " From Home page

Link: https://www.figma.com/proto/VE5fXH6vHOTae5KkDH9uKu/Figmabasics?node-id=601-9&t=ttM6ek5qELCjVA7G-1

3. System Tasks and Design

3.1 System Takes:

In this section, we list the important and frequent tasks of our system which are:

1- Patients can sign up to the system and create their personal account.

2- Patients can login to the system and logout of it when needed.

3- Patients can explore the hospital services through the homepage.

4- Patients are able to book appointments by selecting the clinic, doctor, date, and time through the system.

5- Patients can keep track of their appointments through the system.

6- Patients receive system notifications and reminders about their appointments.

3.2 Design Description

We worked hard to revise and convert our previous low-fidelity prototype into a highfidelity prototype with user-friendly interfaces and a practical clickable interactive system. Our goal was to design a clear and simple application that helps patients manage hospital appointments easily.

The design focuses on simplicity and organization so that users can navigate the system without confusion. The application allows users to explore services, book appointments, check waiting times, and receive reminders. We also ensured that the design is clear and uncomplicated so that it can be easily used by different types of users, especially elderly patients.

In terms of design, we focused on creating a professional healthcare-related interface that reflects trust and reliability. The application uses clear icons, readable text, and simple navigation to make the system comfortable and easy for users.

3.3 Interaction style

In our application we used several interaction styles:

1- Drop down menus:

to allow users to choose options such as clinic, doctor, date, and time when booking appointments.

2- Icons:

to clarify services and make it easier for the user to understand the application such as appointment booking, notifications, and home navigation.

3- Buttons: to perform actions such as booking, editing, or canceling appointments.

4- Pop-up windows:

to confirm actions such as canceling or confirming appointments.

3.4 User improvements

We believe that we have designed an application that will greatly improve the hospital experience for patients. Instead of waiting in long queues or visiting the hospital just to schedule appointments, users can now book their appointments easily through the system.

The application allows patients to check waiting times, receive reminders, and manage their appointments from anywhere. This helps reduce confusion and stress related to hospital visits.

Therefore, the system helps save time and effort for patients while also helping hospital staff manage appointments more efficiently. Overall, the application improves communication between patients and hospital staff and creates a smoother healthcare experience.

4 Usability Evaluation

4.1Our Evaluation

In this evaluation, we analyzed the Heal Hospital prototype based on key usability principles such as learnability, efficiency, consistency, visibility, and error prevention.

The interface is simple and easy to understand, especially for first-time users. The layout is well organized, and the spacing between elements improves readability and reduces visual clutter.

Navigation between screens is smooth and logical. Users can easily move from the home screen to searching for doctors, viewing doctor details, and booking appointments. The system structure supports the main user tasks effectively.

The booking process is straightforward. Users can select a doctor, choose a suitable time, and confirm the appointment in a few steps. This reduces effort and makes the process more efficient.

In addition, clear buttons and consistent design elements help users quickly identify actions. Important information such as appointment details and confirmation messages is clearly displayed.

Overall, the prototype provides a user-friendly and efficient interface that supports both patients and hospital staff

4.2 AI Tool Evaluation

The Heal Hospital prototype was evaluated using the AI tool UX Pilot.

The AI analysis showed that the system has a clean and structured layout, which makes it easy for users to navigate and understand the available functions. The interface supports key tasks such as searching for doctors, booking appointments, and viewing appointment details.

The AI tool also confirmed that the design has good visual clarity and logical navigation, which improves the overall user experience.

However, the AI suggested some improvements, including:

- Adding a search bar to improve doctor search
- Improving spacing between elements to enhance readability
- Adding doctor profile images to support decision-making
- Providing calendar integration for better appointment management

Overall, the AI evaluation confirmed that the system is usable and effective, with some Suggestions for further improvement

4.3 Comparison Table

AI Tool Evaluation	Our Evaluation	Usability Aspect
AI confirmed simplicity and ease of use	Easy for new users to understand and use	Learnability
AI confirmed efficient task flow	Tasks can be completed quickly with few steps	Efficiency
AI confirmed consistent structure	Consistent layout and design across screens	Consistency
AI confirmed clarity but suggested improving spacing	Information is clearly displayed	Visibility
AI suggested improving feedback messages	Clear steps reduce user errors,	Error Prevention
AI suggested adding doctors' profile pictures, providing calendar integration, and a search bar.	System supports main tasks effectively	Improvements

4.4 Conclusion

The usability evaluation of the Heal Hospital prototype shows that the system provides a clear and efficient interface for managing hospital appointments.

Both the manual evaluation and the AI evaluation produced similar results. They confirmed that the system is easy to learn, efficient to use, and visually clear. This indicates that the design meets the usability goals of the system.

In addition, the AI tool suggested several improvements, such as adding search functionality, improving spacing, and integrating calendar features. These suggestions can further enhance the user experience.

Overall, the comparison helped validate the design and identify areas for improvement, which can be considered in future development.

User 1:

Results for User 1:

User 1 was able to perform eight tasks of ten tasks accurately without making any errors. The user was able to complete each of the tasks within an acceptable duration of time; however, the task "Create Account " took the longest time of three minutes while the other tasks required a minimum of 30 seconds to a maximum of 90 seconds.

Only two minor errors were made when performing tasks. One of these errors was identified in the task " Book Appointment," when the user selected the wrong specialty and corrected his mistake. This implies that it would be helpful to make sure that the selected specialty is highlighted. Another error was found in the task "View Details ." The user got confused regarding the position of the doctor's specialty and his name. While the doctor's name should appear beneath the specialty, the doctor's name appeared above the specialty.

Overall, the user performed well regarding usability skills.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Create Account	3 minutes	0	Task was executed perfectly without errors	None
2. Login	1 minute	0	Task was executed perfectly without errors	None
3. Select User type	30 seconds	0	Task was executed perfectly without errors	None
4. Book Appointment	2 minutes	1	The user clicked on the wrong specialty at first,then corrected it	Highlight the selected specialty to make it clearer
5. Cancel appointment	40 seconds	0	Task was executed perfectly without errors	None
6. My appointments	30 seconds	0	Task was executed perfectly without errors	None
7. Contact us	1 minute	0	Task was executed perfectly without errors	None
8. Reschedule	1 minute and 30 seconds	0	Task was executed perfectly without errors	None
9. Notifications	30 seconds	0	Task was executed perfectly without errors	None
10. View details	40 seconds	1	The doctors name appears above the specialty label which is confusing	Move the doctors name below the specialty label for correct information hierarchy

User 2:

Results for User 2:

User 2 was successfully able to complete all but one task without committing any mistakes. All tasks were accomplished by the user within an acceptable timeframe, with "Create Account" requiring the longest time of 4 minutes and "Book Appointment" 3 minutes, while other tasks were completed within 20 seconds to 1 minute.

There was only one mistake made by the user in "My Appointments" where the user became confused between the Upcoming and the Past section of appointments due to the lack of distinction of the white and grey background coloration. To resolve this issue, it is advised to make use of differentiating colors to help users identify Upcoming and Past appointments more easily.

User 2 successfully completed all tasks without encountering too many difficulties, indicating that the design of the system is quite intuitive to use.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1.Create account	4 minutes	0	Task was executed perfectly without errors	None
2.Login	1 minute	0	Task was executed perfectly without errors	None
3.Select User type	20 seconds	0	Task was executed perfectly without errors	None
4.Book Appointment	3 minutes	0	Task was executed perfectly without errors	None
5.Cancel Appointment	1 minute	0	Task was executed perfectly without errors	None
6.My Appointments	1 minute	1	The user got confused between the Upcoming and Past appointments sections as the colour difference between the white and grey backgrounds was not noticeable enough	Use more distinct colors or add clear labels and icons to better differentiate between Upcoming and Past appointments
7.Contact us	40 seconds	0	Task was executed perfectly without errors	None
8.Reschedule	30 seconds	0	Task was executed perfectly without errors	None

9.Notifications	**20 seconds**	**0**	**Task was executed perfectly without errors**	**None**
10.View Details	**1 minute**	**0**	**Task was executed perfectly without errors**	**None**

User 3:

Results for User 3:

User 3 completed most of the tasks successfully without committing any errors. In terms of time, the tasks took about 30 seconds up to 2 minutes, except for "Create Account," which took up to 5 minutes.

Only one error was committed by User 3 during "Create Account" task, where the requirements for the creation of the password were not known to him. This indicates that clearer instructions should be more detailed regarding the formation of passwords.

Overall, the user managed to complete the tasks quickly and easily, indicating that the system is well-designed.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Create Account	**5 minutes**	**1**	**The user was unsure about the password requirements.**	**Add clear instructions for password requirements.**
2. Login	**1 minute and 30 seconds**	**0**	**Task was executed perfectly without errors**	**None**
3. Select User type	**15 seconds**	**0**	**Task was executed perfectly without errors**	**None**
4. Book Appointment	**3 minutes**	**0**	**Task was executed perfectly without errors**	**None**
5. Cancel appointment	**1 minutes**	**0**	**Task was executed perfectly without errors**	**None**
6. My appointments	**15 seconds**	**0**	**Task was executed perfectly without errors**	**None**
7. Contact us	**30 seconds**	**0**	**Task was executed perfectly without errors**	**None**
8. Reschedule	**1minute**	**0**	**Task was executed perfectly without errors**	**None**

9. Notifications	**15 seconds**	**0**	**Task was executed perfectly without errors**	**None**
10. View details	**30 seconds**	**0**	**Task was executed perfectly without errors**	**None**

User 4:

Results for User 4:

User 4 was able to complete most of the tasks successfully with only a few minor mistakes. All tasks were completed within an acceptable time range, between 20 seconds and 1 minute, except for "Create Account" which took 4 minutes and "Book Appointment" which took 3 minutes.

A total of three minor errors were observed. One occurred during "Create Account" due to unclear password requirements. Another error happened in "Book Appointment" when the user selected the wrong specialty initially. The third issue appeared in "View Details", where the user was confused about the arrangement of the doctor's name and specialty.

Overall, the user completed the tasks with minimal difficulties, indicating that the system is generally easy to use, but some improvements in clarity and layout are recommended.

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Create Account	**4 minutes**	**1**	**The user hesitated while entering the password**	**Provide clear password requirements .**
2. Login	**1 minute**	**0**	**Task was completed successfully**	**None**
3.Select User type	**25 seconds**	**0**	**Task was completed easily**	**None**
4.Book Appointment	**3 minutes**	**1**	**The user was confused when choosing a specialty**	**Highlight selected specialty clearly**
5.Cancel appointment	**50 Seconds**	**0**	**Task was completed successfully**	**None**
6.My appointments	**35 seconds**	**0**	**Task was completed easily**	**None**
7.Contact us	**1 minute**	**0**	**The user found the page without issues**	**None**

8.Reschedule	**1 minute**	**0**	**Task was completed successfully**	**None**
9.Notifications	**20 seconds**	**0**	**Task was completed successfully**	**None**
10.View details	**45 seconds**	**1**	**The layout caused slight confusion between doctor name and specialty**	**Improve information hierarchy**

User 5:

Results for User 5:

User 5 was able to perform nine tasks out of ten accurately with only one minor error. The user completed almost all tasks within an acceptable timeframe, ranging from 20 seconds to 60 seconds for most functions. However, the "Create Account" task required the longest duration, taking 4 minutes as the user spent additional time reviewing the interface details.

Only one minor error was observed during the "Book Appointment" task, where the user initially selected an incorrect date before quickly correcting it. This suggests that adding a clearer visual confirmation or a summary before the final booking step would enhance the user experience. Overall, the user's performance was highly efficient, indicating that the system's design is intuitive and supports easy navigation for the majority of the tasks

Task Name	Task Time	Number of Errors	Comments	Suggestion to improve Task Design
1. Create Account	**4 minutes**	**0**	**Task was completed, but the user spent extra time reading the terms.**	**None.**
2. Login	**1 minute**	**0**	**Task was executed perfectly without errors.**	**None**
3.Select User type	**20 seconds**	**0**	**Task was executed perfectly without errors**	**None**
4.Book Appointment	**2 minutes**	**1**	**The user accidentally selected the wrong date and had to re-select**	**Add a visual confirmation or a summary before final booking**

5.Cancel appointment	**50 Seconds**	**0**	**Task was executed perfectly without errors.**	**None**
6.My appointments	**30 seconds**	**0**	**Task was executed perfectly without errors.**	**None**
7.Contact us	**45 minute**	**0**	**Task was completed easily.**	**None**
8.Reschedule	**1 minute**	**0**	**Task was executed perfectly without errors.**	**None**
9.Notifications	**20 seconds**	**0**	**Task was executed perfectly without errors**	**None**
10.View details	**40 seconds**	**0**	**Task was completed successfully**	**None**

Is the system usable?

The first participant completed 80% of the tasks successfully. The second and third participants completed 90% of the tasks, while the fourth participant completed 70%. The fifth participant also completed 90% of the tasks successfully.

According to these results, all participants were able to complete more than 70% of the system tasks successfully. This indicates that the system is usable and easy to use, as most users were able to perform the majority of tasks with minimal issues.

Enhancement:

We will be working on our system and design to improve the overall aspects of Heal Hospital. The design needs to be adjusted based on the feedback gathered from the usability testing sessions to ensure a more seamless experience. Small details matter to us, such as refining the information hierarchy to ensure the doctor's name and specialty are correctly placed, and using more distinct color contrasts to help users differentiate between upcoming and past appointments. Additionally, we aim to provide clearer instructions for password requirements during the account creation process to reduce user hesitation and errors. We will also implement visual highlighting for selected categories, such as medical specialties, to guide the users to accomplish their tasks more accurately and intuitively

Conclusion

The Heal Hospital application is a straightforward tool to solve the issues related to inefficiency of hospital appointments booking. The opportunity to make bookings, edit them, cancel the appointments, get reminders, and view waiting times solves the problem of excessive waiting time and inefficient hospital patient flow. Five usability tests performed by different users show that this application is user-friendly and easy to handle, and the results of the test on the efficiency of AI-based system evaluation prove these points.

The following enhancements may be made in future research on this topic. Integration of the application with a database in a particular hospital, integration of electronic health records may contribute to the more accurate scheduling process. Implementation of telemedicine services can help patients not visit the hospitals in some cases. An AI-based algorithm predicting the queue and the estimated waiting time is another idea. Supporting the application in the Arabic language will increase the number of users from Saudi Arabia.

References :

[1] Agency for Healthcare Research and Quality, “Improving Patient Flow and Reducing Emergency Department Crowding,” Rockville, MD, USA, 2020.

[2] BMC Health Services Research, A. Arain et al., “Effect of reminder systems on patient attendance: A systematic review,” vol. 15, no. 1, 2015.

[3] World Health Organization, “Delivering quality health services: A global imperative for universal health coverage,” Geneva, Switzerland, 2018.

[4] International Journal of Human-Computer Studies, M. Czaja et al., “Factors predicting the use of technology: Findings from the Center for Research and Education on Aging and Technology Enhancement (CREATE),” vol. 68, no. 12, pp. 1375–1389, 2006.

[5] Journal of Medical Internet Research, G. Guy et al., “The effectiveness of reminders on appointment attendance: A meta-analysis,” vol. 14, no. 1, 2012.

[6] Zocdoc, “About Zocdoc: Digital healthcare marketplace,” 2023.